How to balance a chemical equation, step by step

1. Write the equation and draw the **table**: the elements below the equation, and under each formula the number of atoms of each element.

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2. Choose the elements in this order:

metals and other elements | **H** | **O (the last one)**

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3. Count the atoms of the element on the LEFT and on the RIGHT. Are they the same?

↙ ↘

✓ YES

This element is balanced. Keep its coefficients and go on to the **next element** (back to step 3).

✗ NO

Go to the side with **FEWER** atoms of that element and write a coefficient **one unit bigger** in front of the formula that contains it (no number = 1, so we try 2, then 3...).

🔁 **Count again.** Still different? Again, increase by one the coefficient on the side that now has **fewer** atoms, until both sides are equal.

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4. When you change a coefficient, the other elements of that formula change too! Check again ALL the elements you had already balanced.

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5. All the elements are balanced on both sides → **the equation is balanced!** If all the coefficients can be divided by the same number, simplify them.